In [2]:
```
pip install mysql-connector-python
```

```
Requirement already satisfied: mysql-connector-python in c:\users\amogh\appdata\local\programs\python\python312\
lib\site-packages (9.3.0)
Note: you may need to restart the kernel to use updated packages.
[notice] A new release of pip is available: 24.1.1 -> 25.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [3]:
```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Amogh@2003",
    database="air_quality"
)

cursor = conn.cursor()

cursor.execute("SELECT * FROM city_day LIMIT 5")
result = cursor.fetchall()

for row in result:
    print(row)

conn.close()
cursor.close()
```

```
('Ahmedabad', '01-01-2015', '', '', 0.92, 18.22, 17.15, '', 0.92, 27.64, 133.36, 0.0, 0.02, 0.0, '', '')
('Ahmedabad', '02-01-2015', '', '', 0.97, 15.69, 16.46, '', 0.97, 24.55, 34.06, 3.68, 5.5, 3.77, '', '')
('Ahmedabad', '03-01-2015', '', '', 17.4, 19.3, 29.7, '', 17.4, 29.07, 30.7, 6.8, 16.4, 2.25, '', '')
('Ahmedabad', '04-01-2015', '', '', 1.7, 18.48, 17.97, '', 1.7, 18.59, 36.08, 4.43, 10.14, 1.0, '', '')
('Ahmedabad', '05-01-2015', '', '', 22.1, 21.42, 37.76, '', 22.1, 39.33, 39.31, 7.01, 18.89, 2.78, '', '')
```

Out[3]: True

In [4]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

%matplotlib inline
```

In [5]:
```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Amogh@2003",
    database="air_quality"
)

cursor = conn.cursor()
cursor.execute("SHOW TABLES")
tables = cursor.fetchall()

print("Available Tables:")
for table in tables:
    print(table[0])

cursor.close()
conn.close()
```

```
Available Tables:
city_day
city_hour
station_day
station_hour
stations
```

In [6]:
```
import mysql.connector

conn = mysql.connector.connect(
    host="localhost",
    user="root",
    password="Amogh@2003",
    database="air_quality"
)

query = "SELECT * FROM city_day;"
df1= pd.read_sql(query, conn)

conn.close()
```

```python
print(df1.head())

print(f"Dataset contains {df1.shape[0]} rows and {df1.shape[1]} columns")

print(df1.info())

print(df1.describe())
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\134773065.py:11: UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not tested. Please consider using SQLAlchemy.
  df1= pd.read_sql(query, conn)
```

```
     ï»¿City        Date PM2.5 PM10     NO    NO2    NOx NH3     CO    SO2  \
0  Ahmedabad  01-01-2015             0.92  18.22  17.15       0.92  27.64
1  Ahmedabad  02-01-2015             0.97  15.69  16.46       0.97  24.55
2  Ahmedabad  03-01-2015            17.40  19.30  29.70      17.40  29.07
3  Ahmedabad  04-01-2015             1.70  18.48  17.97       1.70  18.59
4  Ahmedabad  05-01-2015            22.10  21.42  37.76      22.10  39.33

       O3  Benzene  Toluene  Xylene AQI AQI_Bucket
0  133.36     0.00     0.02    0.00
1   34.06     3.68     5.50    3.77
2   30.70     6.80    16.40    2.25
3   36.08     4.43    10.14    1.00
4   39.31     7.01    18.89    2.78
Dataset contains 9306 rows and 16 columns
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9306 entries, 0 to 9305
Data columns (total 16 columns):
 #   Column      Non-Null Count  Dtype  
---  ------      --------------  -----  
 0   ï»¿City     9306 non-null   object 
 1   Date        9306 non-null   object 
 2   PM2.5       9306 non-null   object 
 3   PM10        9306 non-null   object 
 4   NO          9306 non-null   float64
 5   NO2         9306 non-null   float64
 6   NOx         9306 non-null   float64
 7   NH3         9306 non-null   object 
 8   CO          9306 non-null   float64
 9   SO2         9306 non-null   float64
 10  O3          9306 non-null   float64
 11  Benzene     9306 non-null   float64
 12  Toluene     9306 non-null   float64
 13  Xylene      9306 non-null   float64
 14  AQI         9306 non-null   object 
 15  AQI_Bucket  9306 non-null   object 
dtypes: float64(9), object(7)
memory usage: 1.1+ MB
None
               NO          NO2          NOx           CO          SO2  \
count  9306.00000  9306.000000  9306.000000  9306.000000  9306.000000
mean     19.28640    35.906632    36.017483     3.924563    18.950298
std      22.12934    28.325000    30.602716    10.785396    23.198765
min       0.06000     0.040000     0.150000     0.000000     0.360000
25%       5.97000    16.002500    16.072500     0.570000     6.640000
50%      11.47000    28.990000    26.625000     0.900000    10.890000
75%      23.31750    46.420000    45.620000     1.570000    19.367500
max     221.41000   277.310000   259.540000   175.810000   186.080000

                O3      Benzene      Toluene       Xylene
count  9306.000000  9306.000000  9306.000000  9306.000000
mean     36.523231     3.651298    12.882549     3.321903
std      21.902552     5.421747    17.327180     6.270897
min       0.020000     0.000000     0.000000     0.000000
25%      20.390000     0.590000     1.980000     0.400000
50%      32.645000     2.140000     6.440000     1.350000
75%      47.727500     4.460000    17.065000     3.820000
max     162.430000    77.430000   271.920000   170.370000
```

In [7]:
```python
len(df1)
```

Out[7]:
```
9306
```

In [8]:
```python
null_counts = df1.isnull().sum()

null_percentage = (df1.isnull().mean() * 100).round(2)

null_report = pd.DataFrame({
    'Null Count': null_counts,
    'Null Percentage (%)': null_percentage
})
```

```
null_report = null_report[null_report['Null Count'] > 0]
print(null_report)
```

```
Empty DataFrame
Columns: [Null Count, Null Percentage (%)]
Index: []
```

In [9]:
```
df1.head()
```

Out[9]:

	ï»¿City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	Ahmedabad	01-01-2015			0.92	18.22	17.15		0.92	27.64	133.36	0.00	0.02	0.00		
1	Ahmedabad	02-01-2015			0.97	15.69	16.46		0.97	24.55	34.06	3.68	5.50	3.77		
2	Ahmedabad	03-01-2015			17.40	19.30	29.70		17.40	29.07	30.70	6.80	16.40	2.25		
3	Ahmedabad	04-01-2015			1.70	18.48	17.97		1.70	18.59	36.08	4.43	10.14	1.00		
4	Ahmedabad	05-01-2015			22.10	21.42	37.76		22.10	39.33	39.31	7.01	18.89	2.78		

In [10]:
```
df1['Date'] = pd.to_datetime(df1['Date'], errors='coerce')
```

In [11]:
```
print(df1.dtypes)
print(df1['Date'].head())
```

```
ï»¿City               object
Date          datetime64[ns]
PM2.5                 object
PM10                  object
NO                   float64
NO2                  float64
NOx                  float64
NH3                   object
CO                   float64
SO2                  float64
O3                   float64
Benzene              float64
Toluene              float64
Xylene               float64
AQI                   object
AQI_Bucket            object
dtype: object
0   2015-01-01
1   2015-02-01
2   2015-03-01
3   2015-04-01
4   2015-05-01
Name: Date, dtype: datetime64[ns]
```

In [12]:
```
print(df1.isnull().sum())
```

```
ï»¿City          0
Date          5584
PM2.5            0
PM10             0
NO               0
NO2              0
NOx              0
NH3              0
CO               0
SO2              0
O3               0
Benzene          0
Toluene          0
Xylene           0
AQI              0
AQI_Bucket       0
dtype: int64
```

In [13]:
```
df1.columns = df1.columns.str.strip()
df1.columns = df1.columns.str.replace('ï»¿City', 'City')
```

In [14]:
```
df1.head()
```

Out[14]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_Bucket
0	Ahmedabad	2015-01-01			0.92	18.22	17.15		0.92	27.64	133.36	0.00	0.02	0.00		
1	Ahmedabad	2015-02-01			0.97	15.69	16.46		0.97	24.55	34.06	3.68	5.50	3.77		
2	Ahmedabad	2015-03-01			17.40	19.30	29.70		17.40	29.07	30.70	6.80	16.40	2.25		
3	Ahmedabad	2015-04-01			1.70	18.48	17.97		1.70	18.59	36.08	4.43	10.14	1.00		
4	Ahmedabad	2015-05-01			22.10	21.42	37.76		22.10	39.33	39.31	7.01	18.89	2.78		

In [15]:
```
df1
```

Out[15]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_E
0	Ahmedabad	2015-01-01			0.92	18.22	17.15		0.92	27.64	133.36	0.00	0.02	0.00		
1	Ahmedabad	2015-02-01			0.97	15.69	16.46		0.97	24.55	34.06	3.68	5.50	3.77		
2	Ahmedabad	2015-03-01			17.40	19.30	29.70		17.40	29.07	30.70	6.80	16.40	2.25		
3	Ahmedabad	2015-04-01			1.70	18.48	17.97		1.70	18.59	36.08	4.43	10.14	1.00		
4	Ahmedabad	2015-05-01			22.10	21.42	37.76		22.10	39.33	39.31	7.01	18.89	2.78		
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9301	Visakhapatnam	NaT	7.63	32.27	5.91	23.27	17.19	11.15	0.46	6.87	19.90	1.45	5.37	1.45	47	
9302	Visakhapatnam	NaT	15.02	50.94	7.68	25.06	19.54	12.47	0.47	8.55	23.30	2.24	12.07	0.73	41	
9303	Visakhapatnam	NaT	24.38	74.09	3.42	26.06	16.53	11.99	0.52	12.72	30.14	0.74	2.21	0.38	70	Satis
9304	Visakhapatnam	NaT	22.91	65.73	3.45	29.53	18.33	10.71	0.48	8.42	30.96	0.01	0.01	0.00	68	Satis
9305	Visakhapatnam	NaT	16.64	49.97	4.05	29.26	18.80	10.03	0.52	9.84	28.30	0.00	0.00	0.00	54	Satis

9306 rows × 16 columns

In [16]:
```python
numerical_cols = df1.select_dtypes(include=['float64', 'int64']).columns
categorical_cols = df1.select_dtypes(include=['object']).columns

for col in numerical_cols:
    df1[col] = df1[col].fillna(df1[col].mean())

for col in categorical_cols:
    if df1[col].isnull().sum() > 0:
        df1[col] = df1[col].fillna(df1[col].mode()[0])

print(df1.isnull().sum())
```

```
City             0
Date          5584
PM2.5            0
PM10             0
NO               0
NO2              0
NOx              0
NH3              0
CO               0
SO2              0
O3               0
Benzene          0
Toluene          0
Xylene           0
AQI              0
AQI_Bucket       0
dtype: int64
```

In [17]:
```python
df1 = df1.dropna(subset=['Date'])

print(df1.isnull().sum())
```

```
City          0
Date          0
PM2.5         0
PM10          0
NO            0
NO2           0
NOx           0
NH3           0
CO            0
SO2           0
O3            0
Benzene       0
Toluene       0
Xylene        0
AQI           0
AQI_Bucket    0
dtype: int64
```

In [18]:

```python
len(df1)
```

Out[18]: 3722

In [19]:

```python
df1
```

Out[19]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	AQI	AQI_
0	Ahmedabad	2015-01-01			0.92	18.22	17.15		0.92	27.64	133.36	0.00	0.02	0.00		
1	Ahmedabad	2015-02-01			0.97	15.69	16.46		0.97	24.55	34.06	3.68	5.50	3.77		
2	Ahmedabad	2015-03-01			17.40	19.30	29.70		17.40	29.07	30.70	6.80	16.40	2.25		
3	Ahmedabad	2015-04-01			1.70	18.48	17.97		1.70	18.59	36.08	4.43	10.14	1.00		
4	Ahmedabad	2015-05-01			22.10	21.42	37.76		22.10	39.33	39.31	7.01	18.89	2.78		
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9283	Visakhapatnam	2020-08-06	31.14	125.15	6.55	41.56	27.26	9.71	0.64	6.61	36.65	3.47	9.76	1.78	115	M
9284	Visakhapatnam	2020-09-06	31.84	85.41	4.69	50.11	30.35	10.01	0.88	5.06	33.86	4.07	9.24	2.52	96	Sati
9285	Visakhapatnam	2020-10-06	32.55	79.28	4.78	44.59	27.38	9.34	0.69	5.90	35.06	2.96	7.89	2.08	92	Sati
9286	Visakhapatnam	2020-11-06	37.71	98.53	9.97	52.10	35.79	9.91	0.70	5.59	26.59	5.29	10.38	3.38	90	Sati
9287	Visakhapatnam	2020-12-06	37.67	76.27	12.17	53.59	35.52	9.85	0.97	5.74	17.35	4.77	18.89	2.85	84	Sati

3722 rows × 16 columns

In [21]:

```python
for col in ['PM2.5', 'PM10', 'AQI']:
    df1[col] = pd.to_numeric(df1[col], errors='coerce')

for col in ['PM2.5', 'PM10', 'AQI']:
    df1[col] = df1[col].fillna(df1[col].mean())

df1['AQI_Bucket'] = df1['AQI_Bucket'].fillna(df1['AQI_Bucket'].mode()[0])

print(df1.isnull().sum())
```

```
City          0
Date          0
PM2.5         0
PM10          0
NO            0
NO2           0
NOx           0
NH3           0
CO            0
SO2           0
O3            0
Benzene       0
Toluene       0
Xylene        0
AQI           0
AQI_Bucket    0
dtype: int64
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\3025642010.py:2: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1[col] = pd.to_numeric(df1[col], errors='coerce')
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\3025642010.py:5: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1[col] = df1[col].fillna(df1[col].mean())
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\3025642010.py:7: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1['AQI_Bucket'] = df1['AQI_Bucket'].fillna(df1['AQI_Bucket'].mode()[0])
```

In [22]:

```
from sklearn.model_selection import train_test_split

X = df1.drop(['AQI', 'AQI_Bucket', 'Date'], axis=1)
y = df1['AQI']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
```

In [23]:

```
print(X_train.shape)
print(X_test.shape)
print(y_train.shape)
print(y_test.shape)
```

```
(2977, 13)
(745, 13)
(2977,)
(745,)
```

In [24]:

```
df1.to_csv('df1.csv', index=False)
```

In [25]:

```
df1.isnull().sum()
```

Out[25]:

```
City          0
Date          0
PM2.5         0
PM10          0
NO            0
NO2           0
NOx           0
NH3           0
CO            0
SO2           0
O3            0
Benzene       0
Toluene       0
Xylene        0
AQI           0
AQI_Bucket    0
dtype: int64
```

In [26]:

```
import numpy as np

df1.replace(r'^\s*$', np.nan, regex=True, inplace=True)

print(df1.isnull().sum())
```

```
City             0
Date             0
PM2.5            0
PM10             0
NO               0
NO2              0
NOx              0
NH3           1167
CO               0
SO2              0
O3               0
Benzene          0
Toluene          0
Xylene           0
AQI              0
AQI_Bucket     189
dtype: int64
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\1334575064.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1.replace(r'^\s*$', np.nan, regex=True, inplace=True)
```

In [28]:
```
df1
```

Out[28]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	
0	Ahmedabad	2015-01-01	72.423147	123.974957	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	19
1	Ahmedabad	2015-02-01	72.423147	123.974957	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	19
2	Ahmedabad	2015-03-01	72.423147	123.974957	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	19
3	Ahmedabad	2015-04-01	72.423147	123.974957	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	19
4	Ahmedabad	2015-05-01	72.423147	123.974957	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	19
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9283	Visakhapatnam	2020-08-06	31.140000	125.150000	6.55	41.56	27.26	9.71	0.64	6.61	36.65	3.47	9.76	1.78	11
9284	Visakhapatnam	2020-09-06	31.840000	85.410000	4.69	50.11	30.35	10.01	0.88	5.06	33.86	4.07	9.24	2.52	9
9285	Visakhapatnam	2020-10-06	32.550000	79.280000	4.78	44.59	27.38	9.34	0.69	5.90	35.06	2.96	7.89	2.08	9
9286	Visakhapatnam	2020-11-06	37.710000	98.530000	9.97	52.10	35.79	9.91	0.70	5.59	26.59	5.29	10.38	3.38	9
9287	Visakhapatnam	2020-12-06	37.670000	76.270000	12.17	53.59	35.52	9.85	0.97	5.74	17.35	4.77	18.89	2.85	8

3722 rows × 16 columns

In [35]:
```
columns_to_fill = df1.select_dtypes(include=['number']).columns.difference(['city','AQI_Bucket'])

df1[columns_to_fill] = df1[columns_to_fill].fillna(df1[columns_to_fill].mean())
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\2668850662.py:3: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1[columns_to_fill] = df1[columns_to_fill].fillna(df1[columns_to_fill].mean())
```

In [36]:
```
df1
```

Out[36]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	
0	Ahmedabad	2015-01-01	72.423147	123.974957	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	19
1	Ahmedabad	2015-02-01	72.423147	123.974957	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	19
2	Ahmedabad	2015-03-01	72.423147	123.974957	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	19
3	Ahmedabad	2015-04-01	72.423147	123.974957	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	19
4	Ahmedabad	2015-05-01	72.423147	123.974957	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	19
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9283	Visakhapatnam	2020-08-06	31.140000	125.150000	6.55	41.56	27.26	9.71	0.64	6.61	36.65	3.47	9.76	1.78	11
9284	Visakhapatnam	2020-09-06	31.840000	85.410000	4.69	50.11	30.35	10.01	0.88	5.06	33.86	4.07	9.24	2.52	9
9285	Visakhapatnam	2020-10-06	32.550000	79.280000	4.78	44.59	27.38	9.34	0.69	5.90	35.06	2.96	7.89	2.08	9
9286	Visakhapatnam	2020-11-06	37.710000	98.530000	9.97	52.10	35.79	9.91	0.70	5.59	26.59	5.29	10.38	3.38	9
9287	Visakhapatnam	2020-12-06	37.670000	76.270000	12.17	53.59	35.52	9.85	0.97	5.74	17.35	4.77	18.89	2.85	8

3722 rows × 16 columns

In [37]:
```
df1[columns_to_fill] = df1[columns_to_fill].fillna(df1[columns_to_fill].mean())
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\2829803998.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1[columns_to_fill] = df1[columns_to_fill].fillna(df1[columns_to_fill].mean())
```

In [38]:
```
df1
```

Out[38]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	
0	Ahmedabad	2015-01-01	72.423147	123.974957	0.92	18.22	17.15	NaN	0.92	27.64	133.36	0.00	0.02	0.00	19
1	Ahmedabad	2015-02-01	72.423147	123.974957	0.97	15.69	16.46	NaN	0.97	24.55	34.06	3.68	5.50	3.77	19
2	Ahmedabad	2015-03-01	72.423147	123.974957	17.40	19.30	29.70	NaN	17.40	29.07	30.70	6.80	16.40	2.25	19
3	Ahmedabad	2015-04-01	72.423147	123.974957	1.70	18.48	17.97	NaN	1.70	18.59	36.08	4.43	10.14	1.00	19
4	Ahmedabad	2015-05-01	72.423147	123.974957	22.10	21.42	37.76	NaN	22.10	39.33	39.31	7.01	18.89	2.78	19
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9283	Visakhapatnam	2020-08-06	31.140000	125.150000	6.55	41.56	27.26	9.71	0.64	6.61	36.65	3.47	9.76	1.78	11
9284	Visakhapatnam	2020-09-06	31.840000	85.410000	4.69	50.11	30.35	10.01	0.88	5.06	33.86	4.07	9.24	2.52	9
9285	Visakhapatnam	2020-10-06	32.550000	79.280000	4.78	44.59	27.38	9.34	0.69	5.90	35.06	2.96	7.89	2.08	9
9286	Visakhapatnam	2020-11-06	37.710000	98.530000	9.97	52.10	35.79	9.91	0.70	5.59	26.59	5.29	10.38	3.38	9
9287	Visakhapatnam	2020-12-06	37.670000	76.270000	12.17	53.59	35.52	9.85	0.97	5.74	17.35	4.77	18.89	2.85	8

3722 rows × 16 columns

In [39]:
```
print(df1['NH3'].unique())
```

```
[nan '10.95' '13.12' ... '9.09' '8.25' '10.01']
```

In [40]:
```
print(df1['NH3'].dtype)
```

```
object
```

In [41]:
```python
df1['NH3'] = pd.to_numeric(df1['NH3'], errors='coerce')
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\3374309359.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1['NH3'] = pd.to_numeric(df1['NH3'], errors='coerce')
```

In [42]:
```python
df1['NH3'].fillna(df1['NH3'].mean(), inplace=True)
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\3624396071.py:1: FutureWarning: A value is trying to be set on
a copy of a DataFrame or Series through chained assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on w
hich we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)'
or df[col] = df[col].method(value) instead, to perform the operation inplace on the original object.


  df1['NH3'].fillna(df1['NH3'].mean(), inplace=True)
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\3624396071.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1['NH3'].fillna(df1['NH3'].mean(), inplace=True)
```

In [43]:
```python
df1
```

Out[43]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene
0	Ahmedabad	2015-01-01	72.423147	123.974957	0.92	18.22	17.15	20.655194	0.92	27.64	133.36	0.00	0.02	0.00
1	Ahmedabad	2015-02-01	72.423147	123.974957	0.97	15.69	16.46	20.655194	0.97	24.55	34.06	3.68	5.50	3.77
2	Ahmedabad	2015-03-01	72.423147	123.974957	17.40	19.30	29.70	20.655194	17.40	29.07	30.70	6.80	16.40	2.25
3	Ahmedabad	2015-04-01	72.423147	123.974957	1.70	18.48	17.97	20.655194	1.70	18.59	36.08	4.43	10.14	1.00
4	Ahmedabad	2015-05-01	72.423147	123.974957	22.10	21.42	37.76	20.655194	22.10	39.33	39.31	7.01	18.89	2.78
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
9283	Visakhapatnam	2020-08-06	31.140000	125.150000	6.55	41.56	27.26	9.710000	0.64	6.61	36.65	3.47	9.76	1.78
9284	Visakhapatnam	2020-09-06	31.840000	85.410000	4.69	50.11	30.35	10.010000	0.88	5.06	33.86	4.07	9.24	2.52
9285	Visakhapatnam	2020-10-06	32.550000	79.280000	4.78	44.59	27.38	9.340000	0.69	5.90	35.06	2.96	7.89	2.08
9286	Visakhapatnam	2020-11-06	37.710000	98.530000	9.97	52.10	35.79	9.910000	0.70	5.59	26.59	5.29	10.38	3.38
9287	Visakhapatnam	2020-12-06	37.670000	76.270000	12.17	53.59	35.52	9.850000	0.97	5.74	17.35	4.77	18.89	2.85

3722 rows × 16 columns

In [44]:
```python
nan_counts = df1.isnull().sum()
print(nan_counts)
```

```
City            0
Date            0
PM2.5           0
PM10            0
NO              0
NO2             0
NOx             0
NH3             0
CO              0
SO2             0
O3              0
Benzene         0
Toluene         0
Xylene          0
AQI             0
AQI_Bucket    189
dtype: int64
```

In [46]:
```python
df1.dropna(inplace=True)
```

```
C:\Users\amogh\AppData\Local\Temp\ipykernel_10832\3016100074.py:1: SettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#retu
rning-a-view-versus-a-copy
  df1.dropna(inplace=True)
```

In [47]:
```python
df1
```

Out[47]:

	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Toluene	Xylene	
30	Ahmedabad	2015-01-02	135.99	123.974957	43.48	42.08	84.57	20.655194	43.48	75.23	102.70	0.40	0.04	25.87	7
31	Ahmedabad	2015-02-02	178.33	123.974957	54.56	35.31	72.80	20.655194	54.56	55.04	107.38	0.46	0.06	35.61	9
32	Ahmedabad	2015-03-02	139.70	123.974957	30.61	28.40	56.73	20.655194	30.61	33.79	73.60	0.17	0.03	11.87	6
33	Ahmedabad	2015-04-02	80.65	123.974957	2.37	22.83	24.00	20.655194	2.37	25.73	47.30	0.00	0.00	0.00	2
34	Ahmedabad	2015-05-02	58.36	123.974957	2.60	21.39	23.31	20.655194	2.60	32.66	53.54	0.00	0.00	0.00	1
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
9283	Visakhapatnam	2020-08-06	31.14	125.150000	6.55	41.56	27.26	9.710000	0.64	6.61	36.65	3.47	9.76	1.78	1
9284	Visakhapatnam	2020-09-06	31.84	85.410000	4.69	50.11	30.35	10.010000	0.88	5.06	33.86	4.07	9.24	2.52	
9285	Visakhapatnam	2020-10-06	32.55	79.280000	4.78	44.59	27.38	9.340000	0.69	5.90	35.06	2.96	7.89	2.08	
9286	Visakhapatnam	2020-11-06	37.71	98.530000	9.97	52.10	35.79	9.910000	0.70	5.59	26.59	5.29	10.38	3.38	
9287	Visakhapatnam	2020-12-06	37.67	76.270000	12.17	53.59	35.52	9.850000	0.97	5.74	17.35	4.77	18.89	2.85	

3533 rows × 16 columns

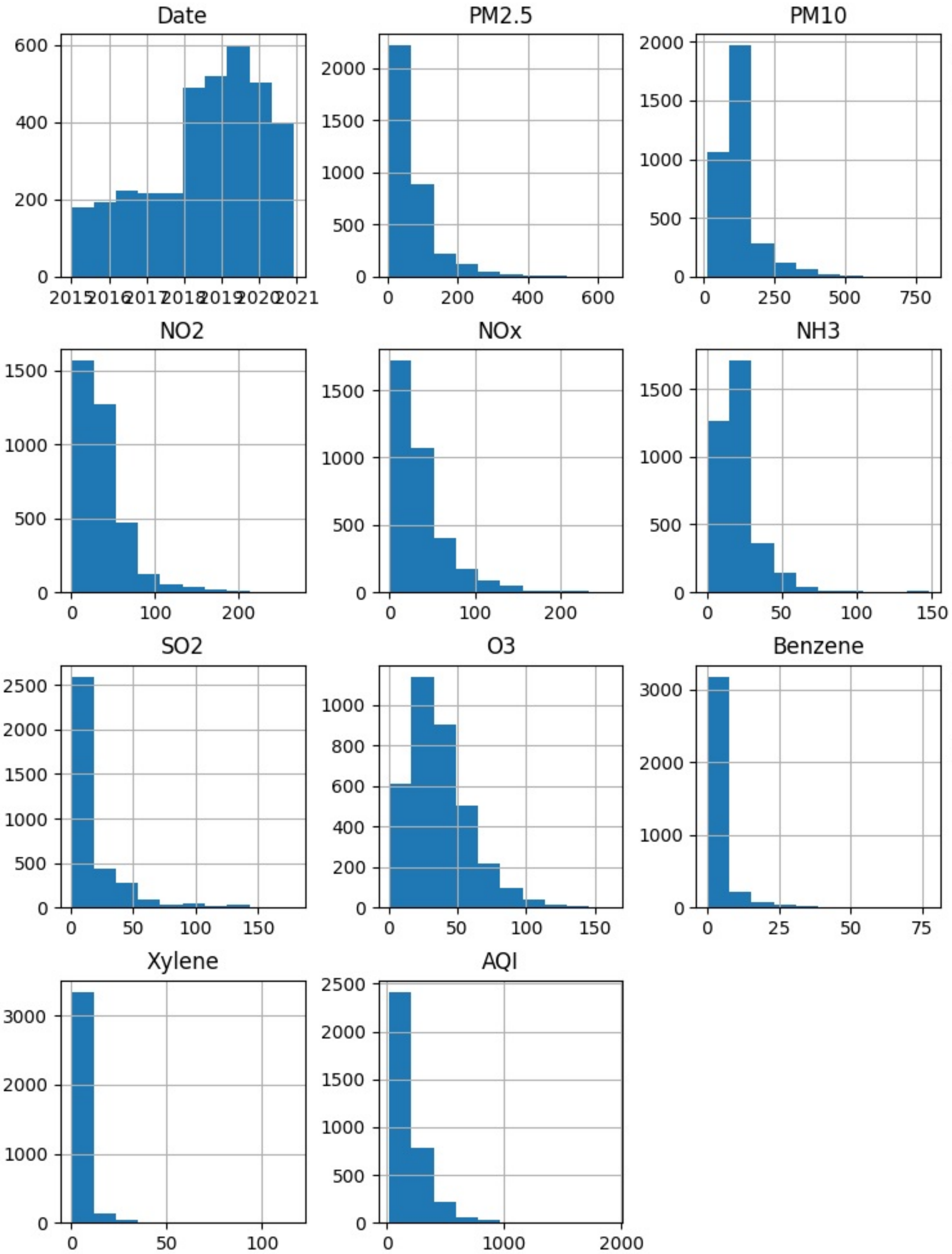
In [48]:
```python
df1.to_csv('df1.csv', index=False)
```

In [49]:
```python
print(df1.describe())
```

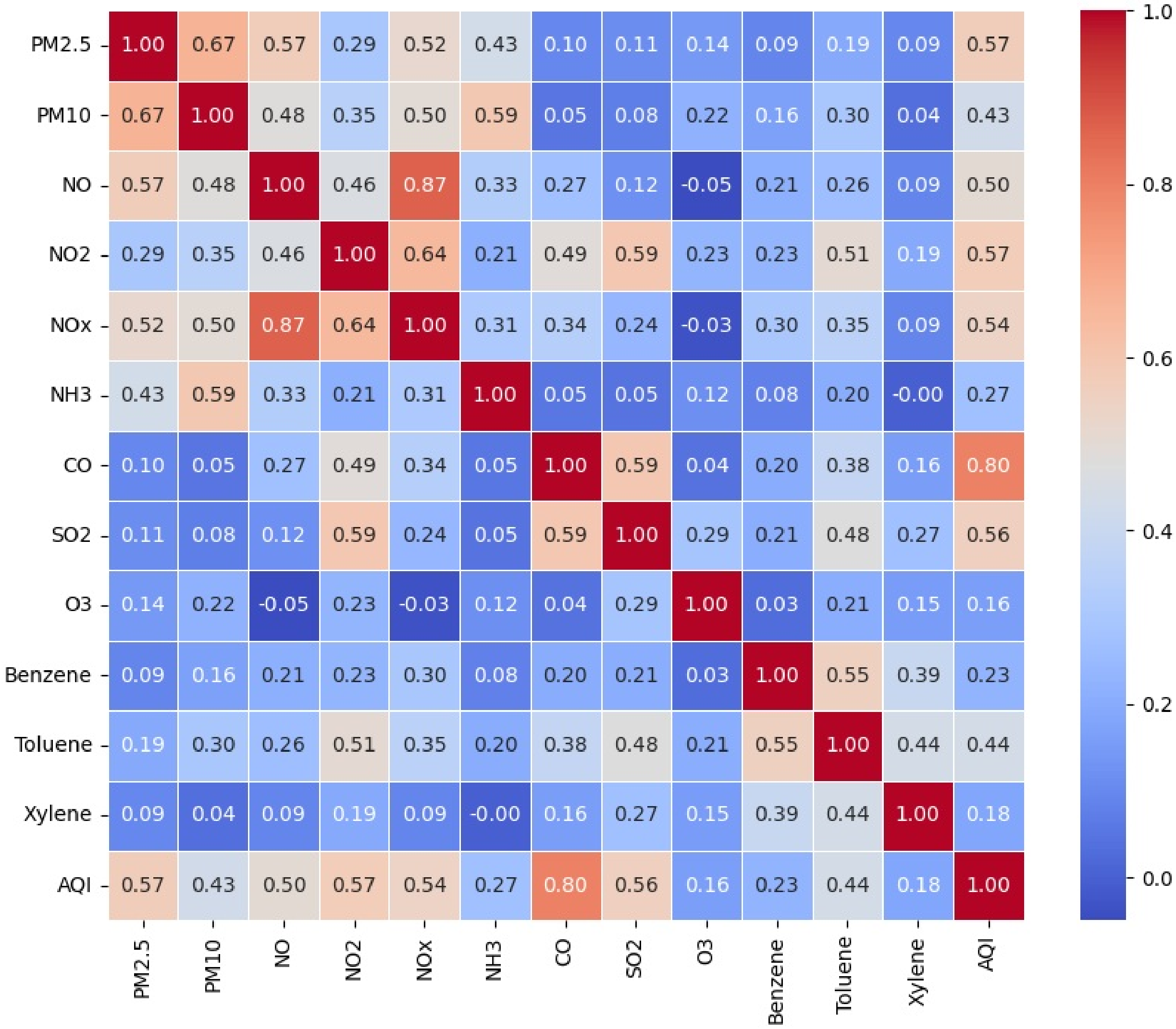
```
                                Date        PM2.5         PM10           NO  \
count                           3533  3533.000000  3533.000000  3533.000000   
mean   2018-07-26 14:03:17.679026176    72.558539   124.030899    19.770628   
min              2015-01-01 00:00:00     4.660000    11.670000     0.400000   
25%              2017-08-04 00:00:00    32.780000    81.330000     6.050000   
50%              2018-11-07 00:00:00    54.050000   123.974957    11.540000   
75%              2019-10-03 00:00:00    85.700000   129.220000    24.330000   
max              2020-12-06 00:00:00   639.190000   796.880000   220.860000   
std                              NaN    65.374230    75.255442    22.529427   

               NO2          NOx          NH3           CO          SO2  \
count  3533.000000  3533.000000  3533.000000  3533.000000  3533.000000   
mean     36.552112    36.424611    20.694252     3.796046    19.038729   
min       0.080000     0.170000     0.120000     0.000000     0.520000   
25%      16.530000    16.220000    12.060000     0.580000     6.740000   
50%      29.710000    26.790000    20.655194     0.920000    10.950000   
75%      47.140000    45.730000    20.655194     1.610000    19.340000   
max     266.460000   259.540000   148.510000   124.010000   178.580000   
std      28.627198    31.360071    13.841604     9.809578    23.118778   

                O3      Benzene      Toluene       Xylene          AQI  
count  3533.000000  3533.000000  3533.000000  3533.000000  3533.000000  
mean     37.018194     3.743156    13.152024     3.439295   196.270592  
min       0.380000     0.000000     0.000000     0.000000    22.000000  
25%      20.630000     0.710000     2.030000     0.400000    91.000000  
50%      33.030000     2.270000     6.650000     1.410000   133.000000  
75%      48.950000     4.730000    17.460000     3.970000   255.000000  
max     162.330000    77.430000   256.090000   116.620000  1917.000000  
std      22.355411     5.286376    17.510975     6.291509   172.508303  
```

In [66]:

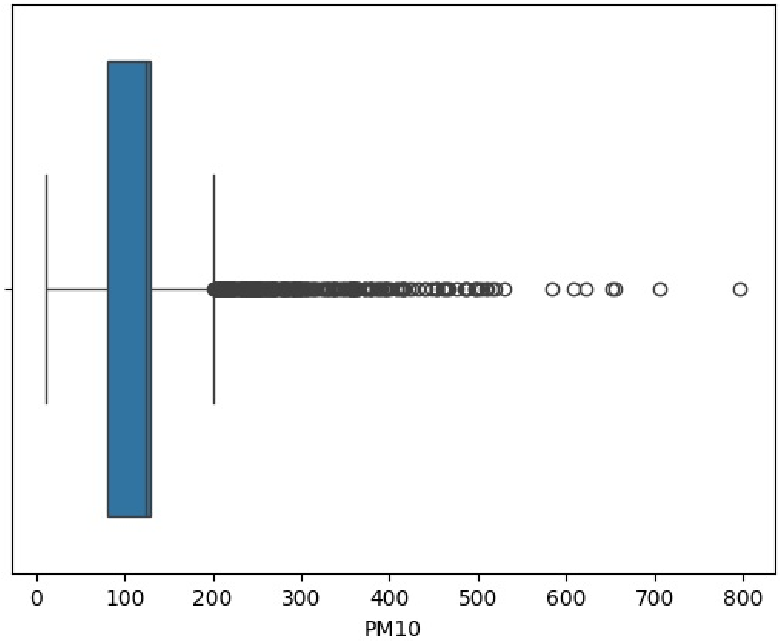
```
import matplotlib.pyplot as plt
df1.hist(figsize=(12, 12))
plt.show()
```



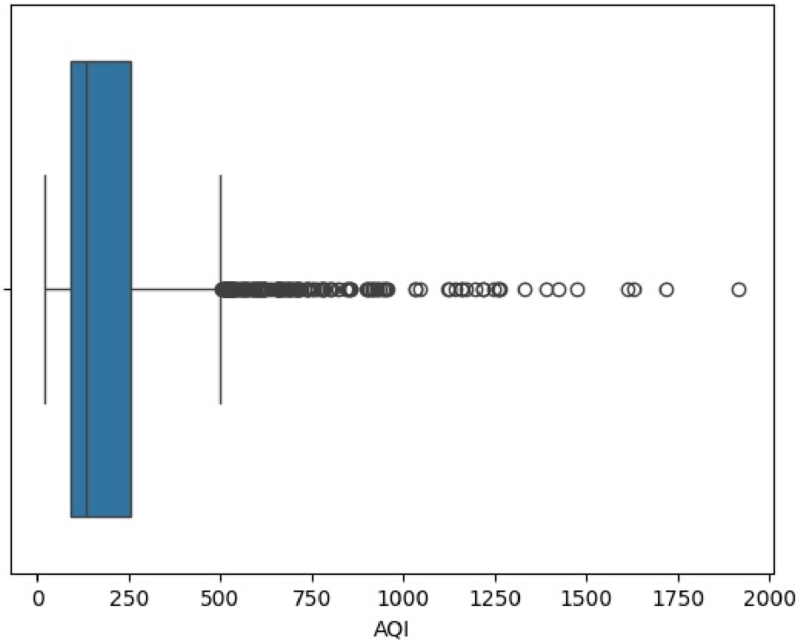
In [68]:
```
import seaborn as sns
import matplotlib.pyplot as plt

numeric_df = df1.select_dtypes(include=['number'])

corr_matrix = numeric_df.corr()

plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.5)
plt.show()
```




```
In [69]: sns.boxplot(x=df1['PM10'])
plt.show()
```




```
In [70]: sns.boxplot(x=df1['AQI'])
plt.show()
```



In [76]:
```
df1.to_csv('df1.csv')
```

In [1]:
```
import pandas as pd
df1=pd.read_csv("df1.csv")
```

In [9]:
```
df1=df1.sort_values(by='AQI', ascending=True)
```

In [10]:
```
df1
```

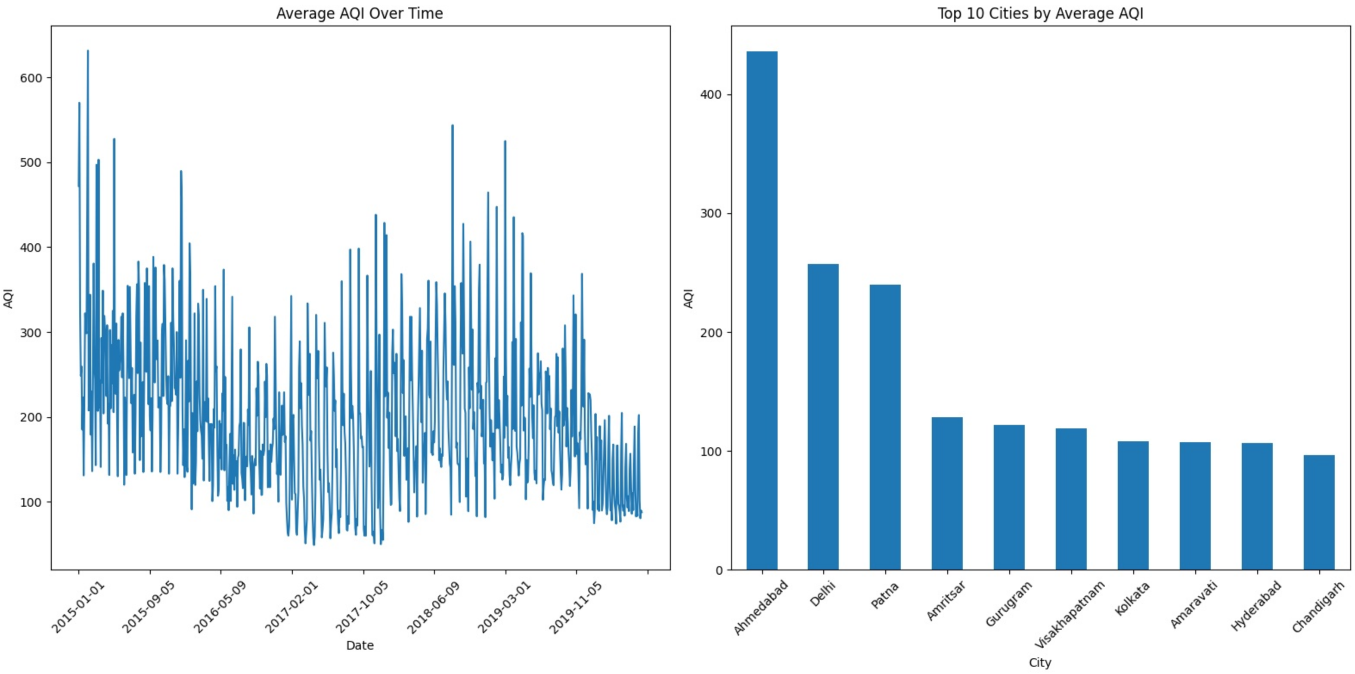
Out[10]:

	Unnamed: 0	City	Date	PM2.5	PM10	NO	NO2	NOx	NH3	CO	SO2	O3	Benzene	Tolue
1806	4746	Hyderabad	2016-06-08	7.29	123.974957	8.99	8.52	17.51	20.655194	0.75	7.02	9.19	0.08	0.
655	1693	Amaravati	2019-03-08	9.31	19.980000	1.48	3.35	2.36	4.400000	0.32	11.83	20.23	0.09	0.
659	1697	Amaravati	2019-08-08	7.45	19.560000	3.47	7.66	6.99	6.310000	0.27	11.94	22.68	0.29	0.
656	1694	Amaravati	2019-05-08	7.53	24.180000	2.74	6.10	5.55	6.380000	0.34	26.08	26.62	0.35	0.
2243	5849	Hyderabad	2019-05-09	11.14	23.810000	3.19	15.09	9.16	9.440000	0.30	7.29	16.22	0.09	0.
...	...	...	...	...	...	...	...	...	...	...	...	...	...	
291	778	Ahmedabad	2019-01-01	110.71	123.974957	63.03	111.56	100.04	20.655194	63.03	80.15	57.12	4.08	32.
182	512	Ahmedabad	2018-01-02	189.12	123.974957	94.00	178.13	160.97	20.655194	94.00	168.44	39.06	0.74	5.
277	729	Ahmedabad	2018-08-11	205.21	123.974957	93.31	100.87	77.22	20.655194	93.31	75.95	61.17	20.41	55.
293	780	Ahmedabad	2019-03-01	131.50	123.974957	119.68	75.82	88.04	20.655194	119.68	55.29	43.25	4.09	32.
191	521	Ahmedabad	2018-10-02	185.77	123.974957	124.01	172.84	186.66	20.655194	124.01	120.94	40.97	0.74	5.

3533 rows × 17 columns

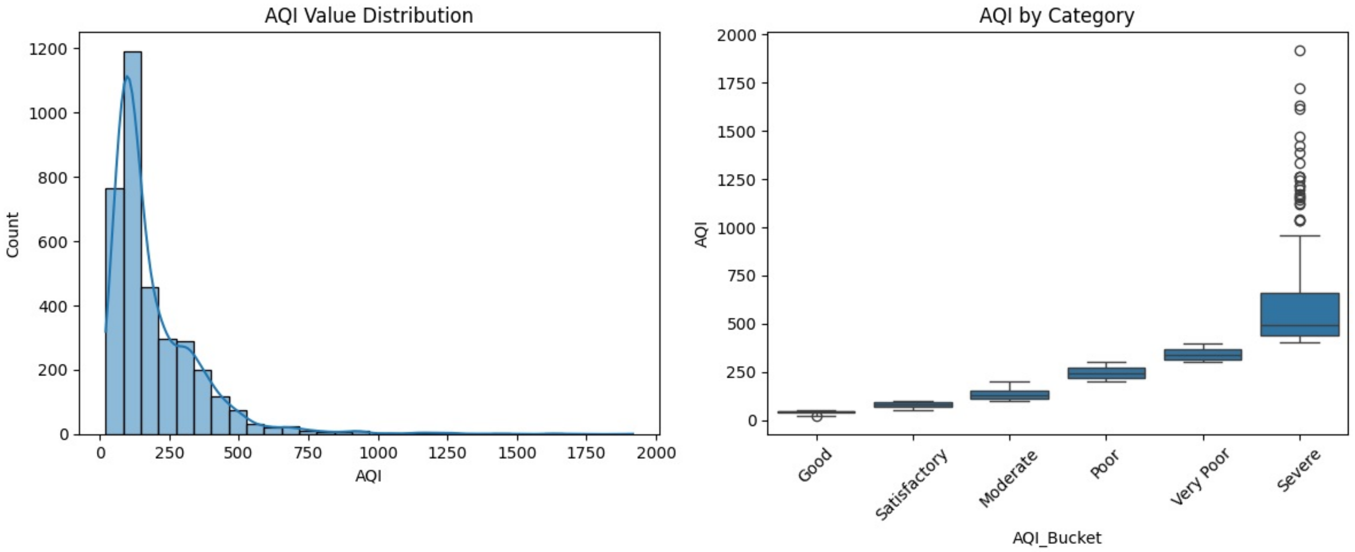
In [23]:
```
import matplotlib.pyplot as plt
import pandas as pd

plt.figure(figsize=(16, 8))

plt.subplot(1, 2, 1)
df1.groupby('Date')['AQI'].mean().plot()
plt.title('Average AQI Over Time')
plt.xlabel('Date')
plt.ylabel('AQI')
plt.xticks(rotation=45)

plt.subplot(1, 2, 2)
```

```python
city_avg = df1.groupby('City')['AQI'].mean().sort_values(ascending=False).head(10)
city_avg.plot(kind='bar')
plt.title('Top 10 Cities by Average AQI')
plt.ylabel('AQI')
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



In [18]:

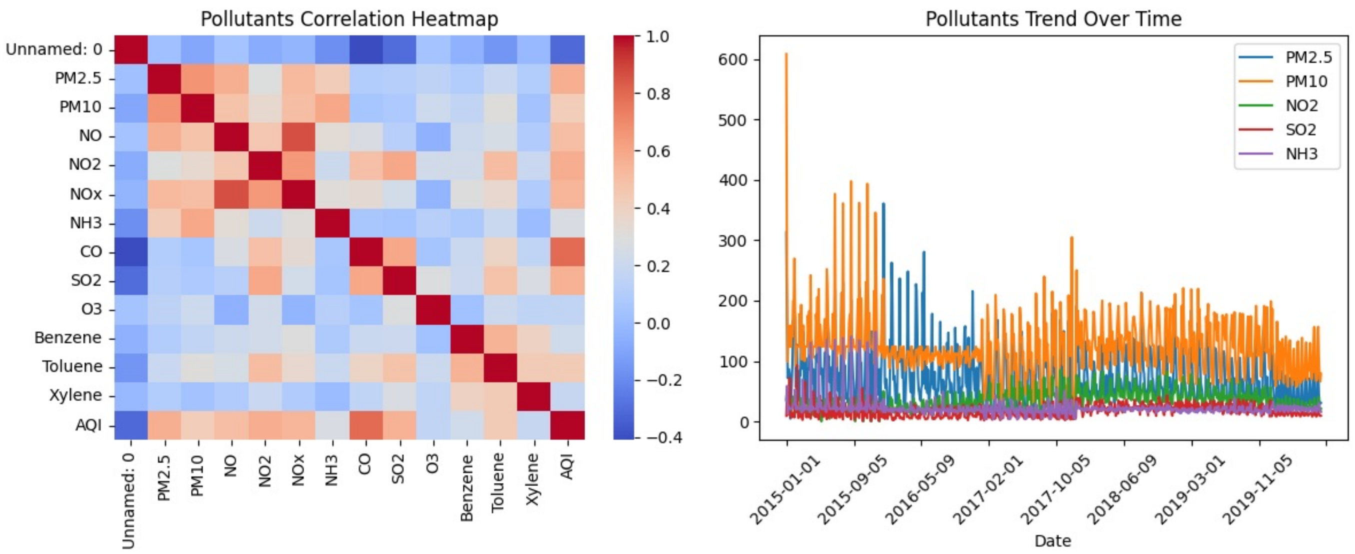
```python
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
sns.histplot(df1['AQI'], bins=30, kde=True)
plt.title('AQI Value Distribution')
plt.xlabel('AQI')

plt.subplot(1, 2, 2)
sns.boxplot(data=df1, x='AQI_Bucket', y='AQI')
plt.title('AQI by Category')
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```



In [22]:

```python
import matplotlib.pyplot as plt
import seaborn as sns

plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
pollutants_corr = df1.select_dtypes(include=['number']).corr()
```

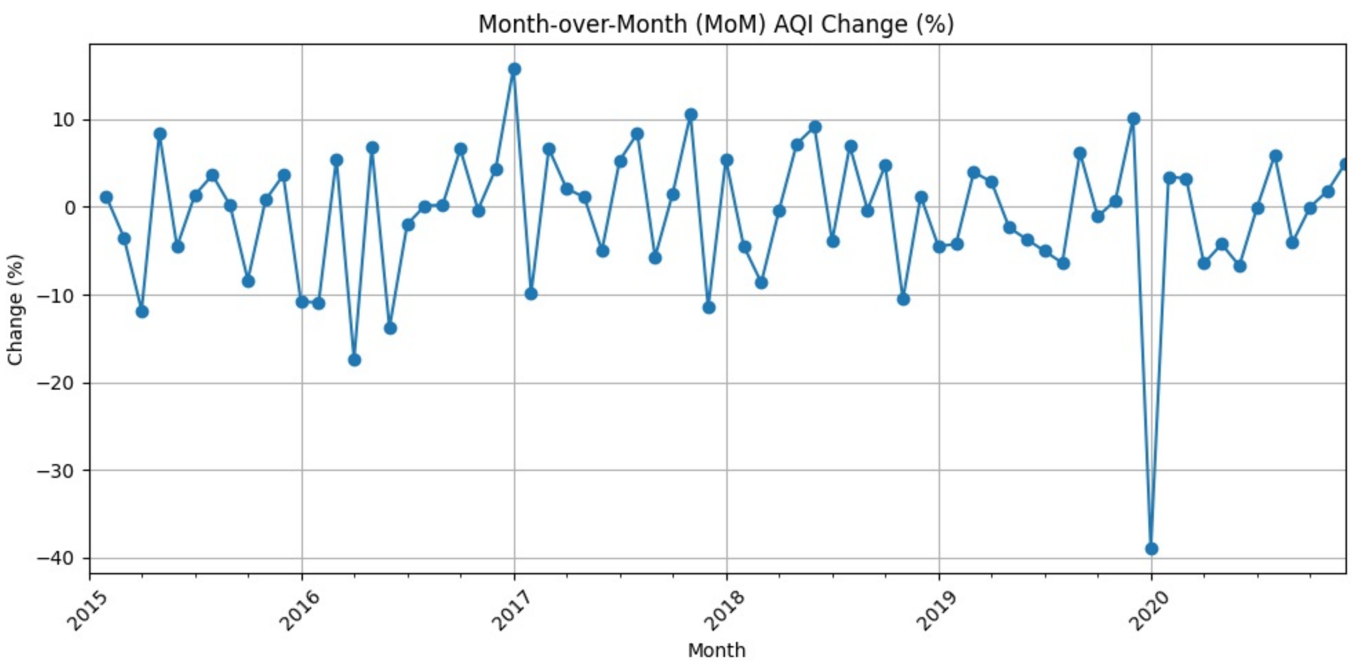
```python
sns.heatmap(pollutants_corr, cmap='coolwarm', annot=False)
plt.title('Pollutants Correlation Heatmap')

plt.subplot(1, 2, 2)
pollutants = ['PM2.5', 'PM10', 'NO2', 'SO2', 'NH3']
df1.groupby('Date')[pollutants].mean().plot(ax=plt.gca())
plt.title('Pollutants Trend Over Time')
plt.xlabel('Date')
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



In [24]:
```python
df1['Date'] = pd.to_datetime(df1['Date'])
```

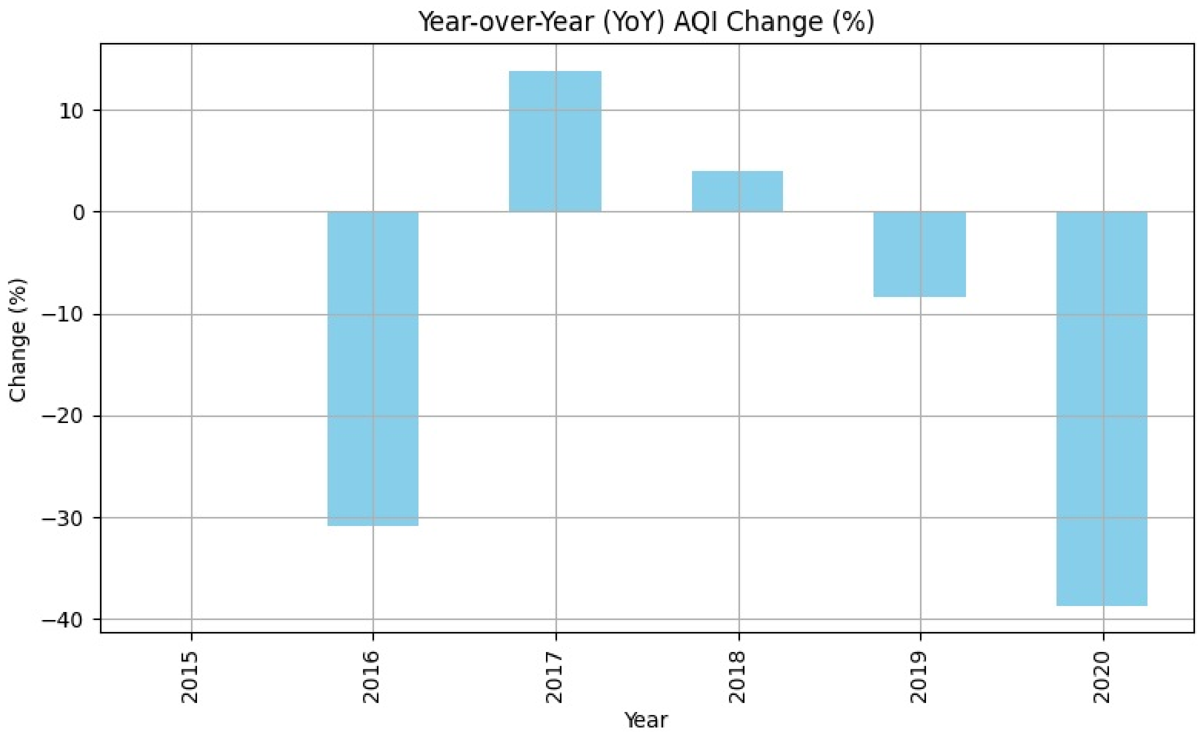
In [25]:
```python
df1['Year'] = df1['Date'].dt.year
df1['Month'] = df1['Date'].dt.to_period('M')
```

In [26]:
```python
monthly_aqi = df1.groupby('Month')['AQI'].mean()

mom_change = monthly_aqi.pct_change() * 100

import matplotlib.pyplot as plt

plt.figure(figsize=(10, 5))
mom_change.plot(marker='o')
plt.title('Month-over-Month (MoM) AQI Change (%)')
plt.xlabel('Month')
plt.ylabel('Change (%)')
plt.xticks(rotation=45)
plt.grid(True)
plt.tight_layout()
plt.show()
```

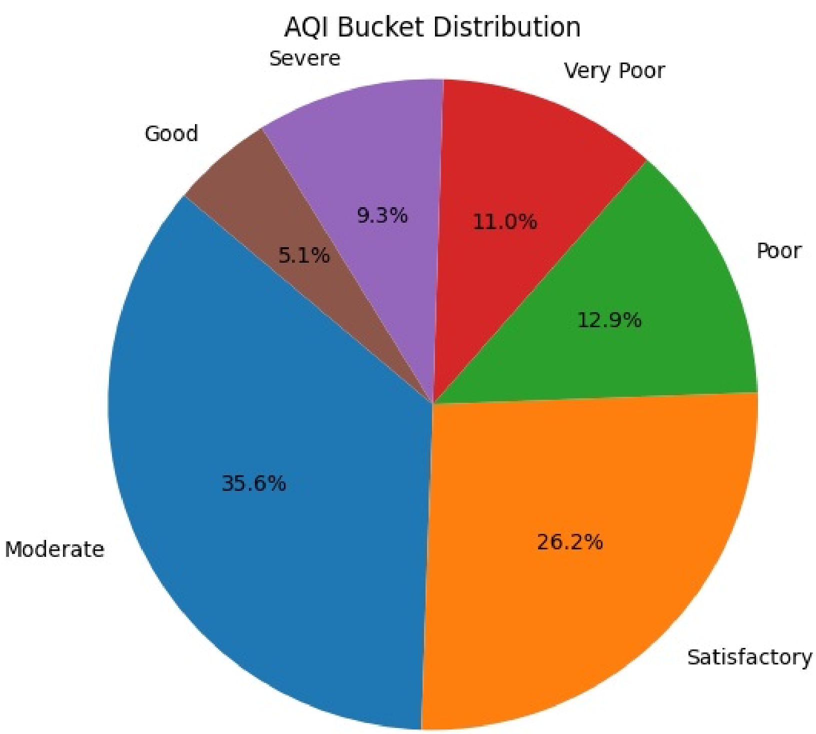


In [27]:
```python
yearly_aqi = df1.groupby('Year')['AQI'].mean()
```

```
yoy_change = yearly_aqi.pct_change() * 100

plt.figure(figsize=(8, 5))
yoy_change.plot(kind='bar', color='skyblue')
plt.title('Year-over-Year (YoY) AQI Change (%)')
plt.xlabel('Year')
plt.ylabel('Change (%)')
plt.grid(True)
plt.tight_layout()
plt.show()
```



In [30]:
```
import matplotlib.pyplot as plt

# Count of each rating category
rating_counts = df1['AQI_Bucket'].value_counts()

# Plot as pie chart
plt.figure(figsize=(6, 6))
plt.pie(rating_counts, labels=rating_counts.index, autopct='%1.1f%%', startangle=140)
plt.title('AQI Bucket Distribution')
plt.axis('equal')
plt.show()
```



Conclusion

To sustain fresh air in the environment, it's essential to monitor and control key pollutants like PM2.5, PM10, NO2, and SO2. Monthly and yearly increases in air quality metrics indicate worsening pollution trends. Key actions include promoting cleaner transportation, enforcing stricter industrial emission standards, and transitioning to renewable energy. Green spaces should be expanded to absorb pollutants, and air quality monitoring should be enhanced for timely interventions. Public awareness campaigns and air quality alerts are crucial for short-term health protection. Long-term sustainability involves continued regulatory improvements and fostering eco-friendly practices. Reducing traffic, limiting industrial emissions, and promoting electric vehicles are vital steps. Ultimately, collective efforts in pollution control and sustainable urban planning are necessary to maintain fresh air.